Memo

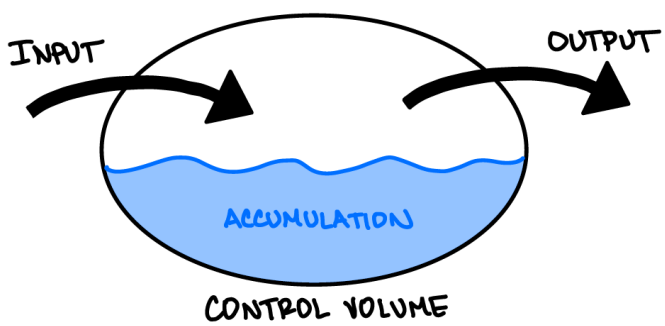
To: [Employee]
From: Anthony Jordan
Date: November 30, 2024
Subject: Reactor Network Transient Response

Dear [Employee],

We have been contracted to support the chemical engineering team at Webb Industries with balancing a system of mixing reactors. Please see the supporting documentation from the client regarding this analysis.

Project Overview:

A plant in north Georgia is comprised of a network of mixing reactors that balance reactant concentrations by controlling mass flow rates between each reactor in the network. A mass balance through a fixed control volume is schematically represented in the following figure:



The modelling interpretation of this figure simply states that the accumulation of mass (assuming an incompressible fluid and no reaction kinetics) per unit of time in a fixed control volume is equal to the difference in the input, or inflow of mass per unit time, and the output, or outflow of mass per unit time:

$$Accumulation = inputs - outputs \left[\frac{kg}{s}\right] \tag{1}$$

Flow into and out of a mixing reactor is often referred to as the mass flow rate, or $\dot{m}$ $[kg/s]$ and can be calculated by multiplying the volumetric flow rate, Q $[m^3/s]$, and the mass concentration, c $[kg/m^3]$, of the stream. Thus equation (1) can be rewritten as:

$$\dot{m}_{accumulation} = \dot{m}_{input} - \dot{m}_{output} \left[\frac{kg}{s}\right] \tag{2}$$

Where each mass flow rate, $\dot{m}$, can be calculated as

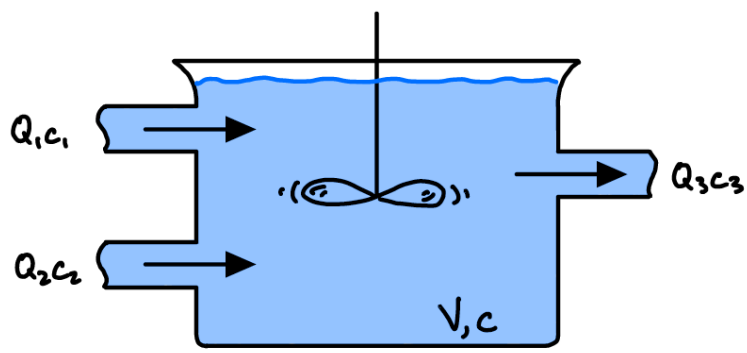
$$\dot{m}\left[\frac{kg}{s}\right] = Q\left[\frac{m^3}{s}\right] \cdot c\left[\frac{kg}{m^3}\right] \tag{3}$$

And the mass flow in/out of a control volume of constant volume, $V\ [m^3]$, and the accumulation can be represented by the following model:

$$V\frac{dc}{dt} = \sum_{i=1}^{n} Q_{in,i}c_{in,i} - \sum_{j=1}^{m} Q_{out,j}c_{out,j} \tag{4}$$

Where the left-hand side represents the accumulation of mass in the system (assuming a constant volume) and the right-hand side represents the net flow into the system minus the net flow out of the system.

For example, consider the following system:



When a mixing reactor first starts, there is a period called the **transient response** where accumulation is a non-zero function of time. Eventually the system needs to stop accumulating or it will lead to system failure, and this period when accumulation has slowed to zero, and thus the flow of mass leaving the system is equal to the flow of mass into the system, is referred to as **steady state**.

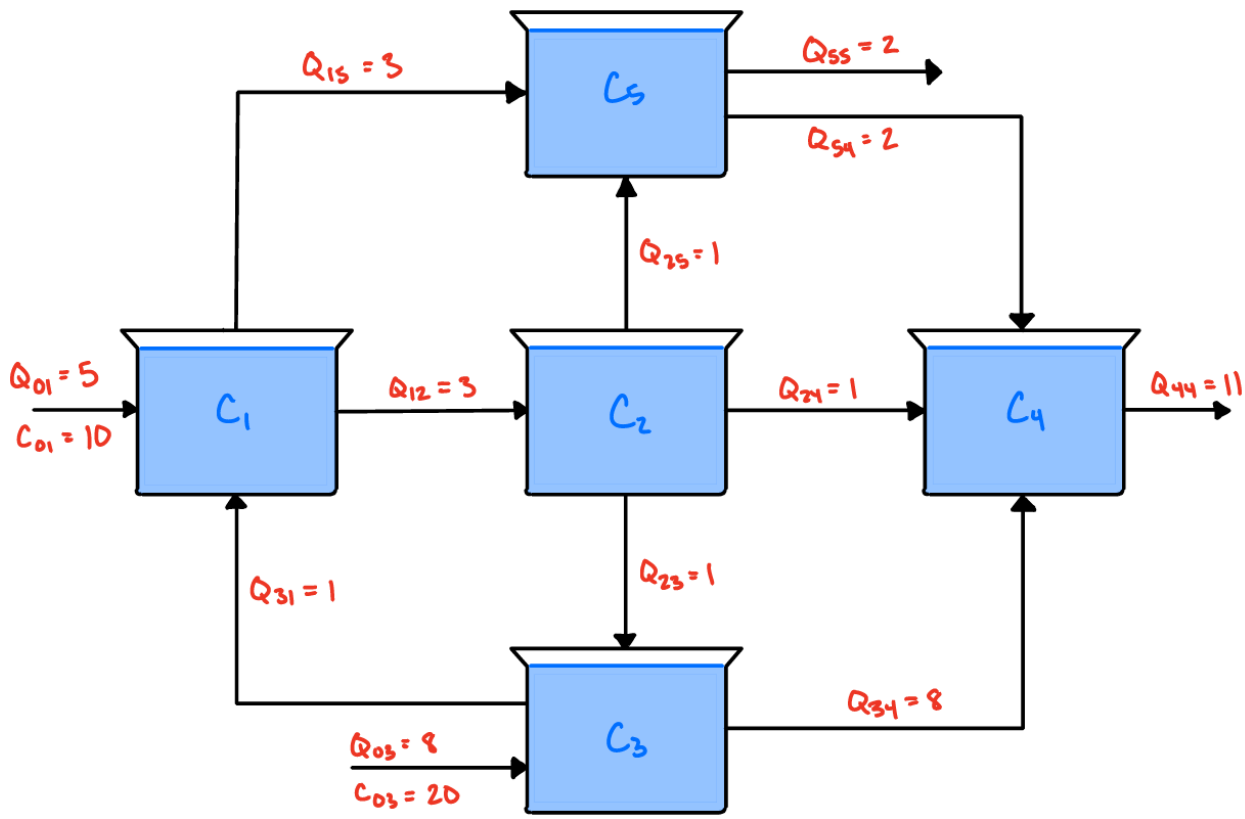
The **transient response** of this system can be modeled using equation (4) as:

$$V\frac{dc}{dt} = Q_1c_1 + Q_2c_2 - Q_3c_3 \tag{5}$$

And in **steady state** the mass balance becomes:

$$Q_1c_1 + Q_2c_2 = Q_3c_3 \tag{6}$$

Webb Industries has many such reactors set up in a large network with the following schematic where all volumetric flow rates are in units of $[m^3/min]$ and all mass concentrations are in units of $[mg/m^3]$:



Keep in mind the following information about the system:

- The reactors have the following volumes:
  - $V_1 = 50\ m^3, V_2 = 20\ m^3, V_3 = 40\ m^3, V_4 = 80\ m^3, V_5 = 100\ m^3$
- Flow and concentration measurement sensors are accurate to 4 significant figures.
- All mixing reactors initially have a concentration of $0\ mg/m^3$.
- Assume perfect mixing in the reactors (there is no distribution of concentration in the reactor – the entire reactor has the same concentration value as it changes).
- All flows except for those labeled with the subscripts 03 and 01 are controlled by flow valves and are thus configurable (relevant for task 3).
- If using **ode45**, passing variables into your **odefun** can be difficult because MATLAB expects it to be formatted with a specific list of input arguments (which cannot easily be modified). While not great coding practice, **`global`** variables might be useful for such a task. There is also supporting documentation in MATLAB help files under "Pass Extra Parameters to ODE Function" here: https://www.mathworks.com/help/matlab/ref/ode45.html

The client has asked for the following information:

- **Steady State Analysis**
  The client wants you to determine the steady state concentrations of each mixing reactor in the system by analyzing each reactor under steady state conditions using mass conservation. You may solve the system of five equations (one for each reactor) in any way

that you see fit.

- **Transient Response Analysis**
  The client then wants you to determine the transient response of each mixing reactor in the system using equation (4), assuming they all start at a concentration of $0\ mg/m^3$. They would like you to perform this analysis using at least a 4th order Runge-Kutta technique (built-in MATLAB solvers are fine) for a time span up to 100 minutes. They would like you to report the rise time, $t_{90}\ [min]$ that it takes each reactor in the system to reach 90% of its steady state value.

- **Mixing Reactor Network Tuning**
  Using your analyses from the previous tasks, the client would like you to suggest a set of volumetric flow rates that will keep all concentrations above $10\ mg/m^3$ in steady state with rise times for all reactors of at most $30\ min$. You may modify any volumetric flow rate governed by a flow control valve. They would like a documented process of your tuning decisions so that they may follow similar logic in the future to further tune their systems. They have no preference for how you document your decisions, but your decisions should be justifiable and defensible.

  ***Breaking the fourth wall* Extra Credit:** I will award up to 5 points of extra credit if you successfully explore, utilize, and sufficiently justify/explain the use of eigenvalues in your tuning decisions. I will not provide you any help here – this is an opportunity for you to do your own exploration and learning to guide your solution. The grading rubric for earning these points will be awarded as follows:
  - +5 – The systematic and correct use of eigenvalues in tuning decisions is well documented and well justified with clear logic and design decisions are clearly communicated to the client in an actionable manner.
  - +0 – The use of eigenvalues was not attempted in tuning logic and decision making.

  There is no partial credit here – either you do this correctly and get the points or you don't. Please understand that this is **in place of** an ad-hoc tuning method and that you will be providing ONE tuning decision-making process to the client. You may not present two different (and possibly conflicting) tuning processes to the client, which means that if you incorrectly implement, explain, and detail this process then you will likely harm your overall score on this project.

  These points will be awarded on top of whatever score you receive. Therefore, you may achieve a possible score of 105/100 on this project.

Deliverables:

- A detailed PowerPoint presentation addressing the tasks above that adheres to client requirements and the Burdell Inc. style guide. See the provided template for reference and

remember to focus primarily on delivering only data and analyses that are actionable for the client's requests. The slides must at least include:

- A .zip file containing your MATLAB scripts and supporting functions that produces all figures and calculations contained in your slides and adhering to the coding guidelines at Burdell Inc. As usual, the entire analysis should be run by executing a single script file and should run in a clean MATLAB environment.

I want you to determine the number of slides and the content of those slides that will make the most compelling case for your conclusions based on the client's needs. Please ensure that your deliverables are clear and concise, as I will be presenting them to the client. Because I haven't performed this analysis myself, I need you to provide detailed presenter notes that explain all your modeling choices, including but not limited to the context under which you are using your numerical frameworks, any relevant information for the client about how they work, and how you analyze the appropriateness and trustworthiness of the results. If you have any questions or need further clarification, do not hesitate to reach out to me.

I look forward to your insightful analysis and contributions to this project.

Best regards,

Anthony Jordan
Engineering Manager
Burdell Inc.